STAT201 · STATISTICS · LECTURE 6

Joint Distributions, Covariance and Correlation

How to describe two random variables at once, and why the relationship between assets determines portfolio risk

Syllabus topic 6 · week 6 · 4 academic hours

LECTURE GOALS

- read and build a joint distribution table and find marginal and conditional distributions;
- check the independence of two discrete variables;
- compute covariance and the correlation coefficient (from a distribution and from a sample) and interpret them;
- find the variance of a sum and a difference of dependent variables;
- compute the risk of a two-asset portfolio and explain the effect of diversification.

1 The joint distribution

So far we have studied a single variable. In economics pairs are interesting: price and demand, the returns of two assets, the sales of two shops. A *joint distribution* specifies the probabilities $p(x, y) = P(X = x,\ Y = y)$ of all pairs of values.

EXAMPLE 1. SALES OF TWO SHOPS

X is the number of units of a product sold per hour in shop A, Y in shop B (values $0, 1, 2$).

	$X = 0$	$X = 1$	$X = 2$	$P(Y = y)$
$Y = 0$	0.10	0.10	0.05	**0.25**
$Y = 1$	0.10	0.30	0.10	**0.50**
$Y = 2$	0.05	0.05	0.15	**0.25**
$P(X = x)$	**0.25**	**0.45**	**0.30**	1

The sum of all nine probabilities is 1. The row and column sums (the margins of the table) give the *marginal* distributions of X and Y.

The marginal distribution is obtained by summation: $p_X(x) = \sum_y p(x, y)$. It gives the usual characteristics: $\mathrm{E}[X] = 0 \cdot 0.25 + 1 \cdot 0.45 + 2 \cdot 0.30 = 1.05$, $\mathrm{Var}(X) = 1.65 - 1.05^2 = 0.5475$; for Y: $\mathrm{E}[Y] = 1$, $\mathrm{Var}(Y) = 0.5$.

1.1 Conditional distributions and independence

The conditional distribution is obtained as in Lecture 3: $p(y \mid x) = p(x,y)/p_X(x)$. For $X = 2$ (the column is divided by 0.30): $P(Y=0 \mid X=2) = 0.1\overline{6}$, $P(Y=1 \mid X=2) = 0.\overline{3}$, $P(Y=2 \mid X=2) = 0.5$. The conditional expectation is $\mathrm{E}[Y \mid X=2] = 0.333 + 1.0 = 1.33$, whereas $\mathrm{E}[Y] = 1$: when shop A sells a lot, shop B on average does too.

INDEPENDENCE

Discrete variables are independent if $p(x,y) = p_X(x)\,p_Y(y)$ for *all* pairs (x,y).

In our example $p(0,0) = 0.10 \neq 0.25 \cdot 0.25 = 0.0625$, so X and Y are dependent. One violation is enough.

2 Covariance and correlation

COVARIANCE

$$\mathrm{Cov}(X,Y) = \mathrm{E}[(X-\mu_X)(Y-\mu_Y)] = \mathrm{E}[XY] - \mu_X\mu_Y, \qquad \mathrm{E}[XY] = \sum_{x,y} xy\,p(x,y). \quad (2.1)$$

A positive covariance means that large values of X are on average accompanied by large values of Y; a negative one means the opposite. Its value depends on the units of measurement, so a dimensionless coefficient is used to assess the strength of the relationship.

CORRELATION

$$\rho = \mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}, \qquad -1 \le \rho \le 1. \quad (2.2)$$

EXAMPLE 2. CONTINUATION: SALES OF TWO SHOPS

Nonzero terms in $\mathrm{E}[XY]$ come only from $x, y \ge 1$: $1\cdot 1\cdot 0.30 + 2\cdot 1\cdot 0.10 + 1\cdot 2\cdot 0.05 + 2\cdot 2\cdot 0.15 = 0.30 + 0.20 + 0.10 + 0.60 = 1.20$. Therefore $\mathrm{Cov}(X,Y) = 1.20 - 1.05\cdot 1 = 0.15$, $\rho = \dfrac{0.15}{\sqrt{0.5475\cdot 0.5}} = 0.287$ — the relationship is positive and weak.

Properties: $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$; $\mathrm{Cov}(aX+b,\, cY+d) = ac\ \mathrm{Cov}(X,Y)$; $\mathrm{Corr}(aX+b,\, cY+d) = \mathrm{sign}(ac)\ \mathrm{Corr}(X,Y)$ (correlation does not depend on units). If X and Y are independent, then $\mathrm{Cov} = 0$ and $\rho = 0$.

COMMON MISTAKE

The converse is false: $\rho = 0$ does not mean independence. Correlation measures only a *linear* relationship. Example: X takes the values $-2, -1, 0, 1, 2$ with equal probability, and $Y = X^2$; Y is completely determined by X, but $\mathrm{Cov}(X,Y) = 0$.

2.1 Sample covariance and correlation

From observations (x_i, y_i), $i = 1..n$:

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}, \qquad r = \frac{s_{xy}}{s_x s_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}. \tag{2.3}$$

EXAMPLE 3. ADVERTISING AND SALES

Six months: advertising spend x (million UZS) $= 2, 3, 5, 6, 8, 10$; sales y (million UZS) $= 15, 17, 22, 31, 33, 40$. $\bar{x} = 5.667$, $\bar{y} = 26.333$. $\sum (x_i - \bar{x})^2 = 45.33$, $\sum (y_i - \bar{y})^2 = 487.33$, $\sum (x_i - \bar{x})(y_i - \bar{y}) = 145.67$ (terms $41.56;\ 24.89;\ 2.89;\ 1.56;\ 15.56;\ 59.22$). Hence $s_{xy} = 145.67/5 = 29.13$, $s_x = 3.01$, $s_y = 9.87$, $r = \dfrac{145.67}{\sqrt{45.33 \cdot 487.33}} = 0.980$.

REMEMBER

Correlation is not causation. A high r may arise because of a third factor, because of reverse causation, or by chance. We discuss this in detail in Lectures 12–13.

3 The sum and difference of dependent variables

Variance of a sum

$$\operatorname{Var}(X \pm Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) \pm 2 \operatorname{Cov}(X, Y).$$

For independent variables the earlier rule remains (Lecture 4). In the shops example: $\operatorname{Var}(X + Y) = 0.5475 + 0.5 + 2 \cdot 0.15 = 1.3475$, and $\operatorname{Var}(X - Y) = 1.0475 - 0.30 = 0.7475$. The expectation is always $\mathrm{E}[X + Y] = 2.05$. A positive relationship makes the sum more variable and the difference less so.

4 A portfolio of two assets

The asset returns are R_1, R_2 with expectations μ_1, μ_2, standard deviations σ_1, σ_2 and correlation ρ. A share w of capital is invested in the first asset, $1 - w$ in the second. The portfolio return is $R_p = wR_1 + (1-w)R_2$:

$$\mathrm{E}\big[R_p\big] = w\mu_1 + (1-w)\mu_2, \qquad \sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\,\sigma_1\sigma_2. \tag{4.1}$$

EXAMPLE 4. DIVERSIFICATION

$\sigma_1 = 10\%$, $\sigma_2 = 15\%$, $w = 0.5$. The standard deviation of the portfolio:

ρ	1	0.5	0	−0.5	−1
σ_p, %	12.50	10.90	9.01	6.61	2.50

Only at $\rho = 1$ does the portfolio risk equal the average risk of the assets (12.5%). The lower the correlation, the stronger the diversification effect; the expected return does not depend on ρ.

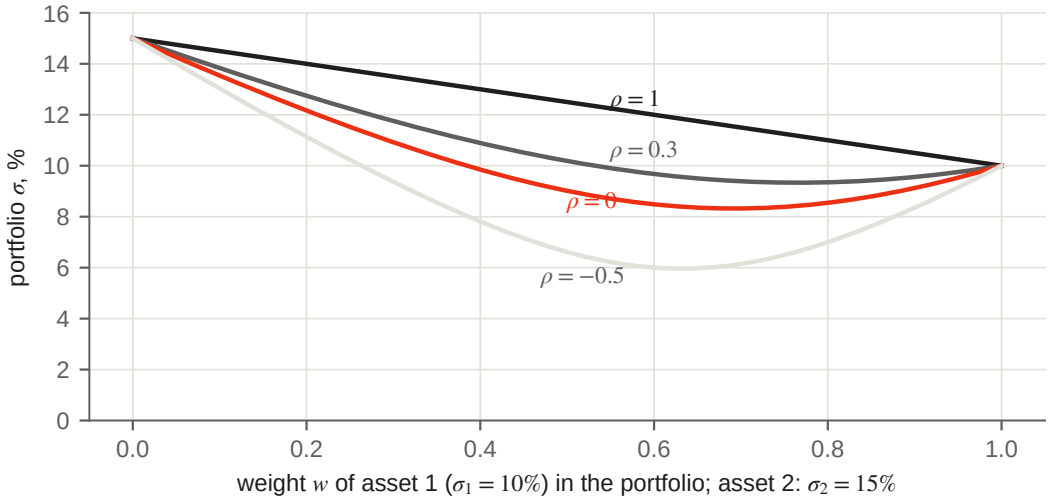


Fig. 1. Portfolio risk as a function of the weight of asset 1 for different correlations. The lower ρ, the deeper the "dip" of the curve.

EXAMPLE 5. MINIMUM RISK

For $\rho = 0.3$ the weight of asset 1 that minimizes the variance is $w^* = \dfrac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} = \dfrac{0.0225 - 0.0045}{0.01 + 0.0225 - 0.009} = 0.766$, $\sigma_p = 9.33\%$ — less than either asset alone (10% and 15%).

The same reasoning explains why the risk of a portfolio of n equally risky loans with pairwise correlation ρ does not fall to zero: $\sigma_{\bar{X}} = \sigma\sqrt{\frac{1+(n-1)\rho}{n}}$. For $\sigma = 2.0$, $n = 100$: at $\rho = 0$ we get 0.20, at $\rho = 0.1$ already 0.66. Systematic (common) risk is not eliminated by diversification.

5 Key terms

Term	*In Russian*	Meaning
Joint distribution	*Совместное распределение*	$p(x,y) = P(X=x, Y=y)$
Marginal distribution	*Маргинальное распределение*	$p_X(x) = \sum_y p(x,y)$
Conditional distribution	*Условное распределение*	$p(y \mid x) = p(x,y)/p_X(x)$
Covariance	*Ковариация*	$\mathrm{E}[XY] - \mu_X\mu_Y$
Correlation	*Корреляция*	$\rho = \mathrm{Cov}/(\sigma_X\sigma_Y)$
Diversification	*Диверсификация*	risk reduction thanks to imperfect correlation

Formulas of Lecture 6

$$\mathrm{Cov}(X,Y) = \mathrm{E}[XY] - \mu_X\mu_Y, \qquad \rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y},$$

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}},$$

$$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\,\mathrm{Cov}(X,Y),$$

$$\mathrm{Cov}(aX + b, cY + d) = ac\,\mathrm{Cov}(X,Y),$$

$$\sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2.$$

6 Practice problems

Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] The joint distribution of two indicators: $P(X=0, Y=0) = 0.3$, $P(X=0, Y=1) = 0.2$, $P(X=1, Y=0) = 0.1$, $P(X=1, Y=1) = 0.4$.

(a) Find the marginal distributions of X and Y.
(b) Find $\mathrm{Cov}(X,Y)$ and ρ.
(c) Find $P(Y=1 \mid X=1)$. Are X and Y independent?

Problem 2 [A] $\mathrm{Var}(X) = 9$, $\mathrm{Var}(Y) = 16$, $\mathrm{Cov}(X,Y) = 6$. Find ρ, $\mathrm{Var}(X+Y)$, $\mathrm{Var}(X-Y)$, $\mathrm{Var}(2X - 3Y)$.

Problem 3 [A] Check independence in two tables (rows $Y = 0, 1$; columns $X = 0, 1$):

(1) $p(0,0)=0.2$, $p(1,0)=0.3$, $p(0,1)=0.2$, $p(1,1)=0.3$;
(2) $p(0,0)=0.1$, $p(1,0)=0.4$, $p(0,1)=0.3$, $p(1,1)=0.2$.
(The first index is the value of X, the second the value of Y.)

Problem 4 [B] Five students: study time x (h) $=1,2,3,4,5$ and score $y=50,55,65,70,85$.

(a) Find $\bar{x}$, $\bar{y}$ and $\sum(x_i-\bar{x})(y_i-\bar{y})$.
(b) Find the sample covariance and correlation.

Problem 5 [B] Assets: $\mu_1=9\%$, $\sigma_1=12\%$; $\mu_2=14\%$, $\sigma_2=20\%$; $\rho=0.2$. Portfolio: 40% in asset 1 and 60% in asset 2.

(a) Find the expected return and standard deviation of the portfolio.
(b) Compare σ_p with the weighted average of the standard deviations. What does the difference show?

Problem 6 [B] $\sigma_1=10\%$, $\sigma_2=15\%$, the assets are uncorrelated ($\rho=0$). At what weight w of asset 1 is the portfolio variance minimal? What is σ_p?

Problem 7 [B] For the data of Problem 4, study time was measured not in hours but in minutes ($x'=60x$), and scores as a fraction of the maximum (suppose the maximum is 100, $y'=y/100$). How do s_{xy} and r change? Compute.

Problem 8 [B] $X\in\{0,1,2\}$ is the number of complaints per week, $Y\in\{1,2\}$ the shift. Joint probabilities $p(x,y)$: $p(0,1)=0.1$, $p(1,1)=0.2$, $p(2,1)=0.1$, $p(0,2)=0.2$, $p(1,2)=0.3$, $p(2,2)=0.1$.

(a) Find the marginal distributions and $\mathrm{E}[X]$, $\mathrm{E}[Y]$.
(b) Find $\mathrm{Cov}(X,Y)$ and ρ.
(c) Find the conditional distribution of X given $Y=2$ and $\mathrm{E}[X\mid Y=2]$.

Problem 9 [B] The costs of two components: $\sigma_1=4$, $\sigma_2=6$. Find the standard deviation of the total cost for $\rho=0.5$, $\rho=0$ and $\rho=-0.5$.

Problem 10 [B] A bank issued 50 loans, each with $\sigma=2.0$ million. Find the standard deviation of the average profit for pairwise correlation $\rho=0$ and $\rho=0.2$; find the limit as $n\to\infty$ for $\rho=0.2$.

Problem 11 [C] Prove that (a) $\mathrm{Cov}(X,Y)=\mathrm{E}[XY]-\mu_X\mu_Y$; (b) $\mathrm{Var}(X+Y)=\mathrm{Var}(X)+\mathrm{Var}(Y)+2\,\mathrm{Cov}(X,Y)$.

Problem 12 [C] Let $Y=aX+b$, $a\neq 0$. Show that $\rho_{XY}=+1$ for $a>0$ and -1 for $a<0$.

Problem 13 [C] X takes the values $-1,0,1$ with probabilities $\frac14,\frac12,\frac14$, and $Y=X^2$. Find $\mathrm{Cov}(X,Y)$ and prove that X and Y are dependent.